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Signals of resolve or a shortcut? Border and air space violations in the Baltic Sea region

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ABSTRACT
The Baltic Sea region regularly experiences low-level military incidents and violations of territorial borders. Both international media and the Baltic elite interpret Russian violations of airspace as intentional and provocative. Based on reports in the *Baltic News Service* on Russian airspace violations, this article assesses whether there are systematic patterns regarding geographical distribution as well as variation over time. Moreover, this article examines whether Russian airspace violations can serve as costly signals of resolve and as a deterrent to further NATO expansion.

KEYWORDS Baltic Sea region; deterrence; foreign policy signaling; airspace violations; border violations; military incidents; russia; nato

The Baltic Sea and its littoral states have a long history of active commercial and cultural exchanges and are strongly integrated in economic and political terms (North 2011). Except for Russia, all states are, after several rounds of accession, now members of the European Union. Cooperation takes place at several levels including the military sector, for example, within the EU's common security and defense policy (Archer 2008), and for some time even including Russia as part of NATO's framework of Partnership for Peace. The region is, however, also known as a hotspot of tensions where, for most part of the twentieth century, the two superpowers' spheres of interests directly met. The Cold War system characterized by the East–West confrontation and the salience of military power (Snow 1995) has dominated the region around the Baltic Sea for many years and left its marks in terms of a high perception of threat on both sides. After the demise of the Soviet Union and the Warsaw Pact, the Baltic and east European states continued to feel threatened by Russia, the dominant power in the region. These countries sought protection by striving to join the security umbrella of the Western alliance.

Despite their membership in NATO, the threat perception is still high among the east European and Baltic states. According to the former Estonian Foreign Minister Marina Kaljurand, for example, 'The question of how to respond to increasing Russian aggression remains Estonia's top priority. The facts are stark: We have seen violations of our airspace and irregular flights and mysterious submarines in the Baltic Sea' (BNS 2016b). The recent report on the security situation in Estonia by the Estonian Foreign

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Intelligence Service (2013) also places the Russian military threat at the top of their agenda, as do the Ministry of National Defense and State Security Department of the Republic of Lithuania's (2020) National Threat Assessment and the Latvian National Security Concept (2019). In turn, Russia feels threatened by NATO, which following the requests of several former Soviet republics and allies, expanded eastwards. After the Ukrainian crisis in 2014, Russia's new military doctrine emphasized NATO's expansion as the main external danger to Russia's security (Konyshev and Sergunin 2018). The situation in the Baltic Sea region is often described as a typical security dilemma where Russia reacts to increased NATO activity and deployments with greater military action (Jakniunaite 2016). Mounting tensions with NATO thus offer one possible explanation for why Russia becomes increasingly involved in deterrent behavior and why more military activity in the region can be observed.

Relations between NATO and Russia, however, do not escalate in a linear fashion toward military crisis escalation. In the early 1990s, while the region's security architecture was still uncertain and demilitarization and border issues with Russia not yet resolved (Blank 1997), NATO's relations in the region, described by some as a 'romantic period' (Forsberg and Herd 2015, 42), evolved toward a strategic partnership and Russia joined the Partnership for Peace program. This short period of cooperation ended, however, in 2003–2004 with the Color Revolutions in Georgia and Ukraine and NATO enlargement to include the Baltic states. While most military and political cooperation between NATO and Russia came to a halt, a few joint anti-terrorist military exercises still took place. In particular, the Ukraine crisis and the annexation of Crimea heaped pressure on NATO to reassure its Baltic members, for example, through an increase in surveillance flights (Forsberg and Herd 2015).

In recent years, there have been reports of increased airspace violations in the Baltic Sea region that have not been thoroughly analyzed to date. Reports of Russian fighter jets and other military planes violating Baltic airspace and of NATO jets scrambling are plentiful. A prominent example is an article in *The New York Times* of an incident that involved President Putin's plane on his way to a summit meeting in Finland in summer 2018, which flew without clearance through Estonian airspace over Vaindloo, an island that is the frequent site of Russian airspace violations. It is up to debate whether the Estonian island is a 'hot spot in the aerial game of chicken between Russia and NATO' and the intrusions a political signal or merely 'just to take a short-cut' (New York Times 2018).

The following assessment will systematically investigate the reports of violations of territorial airspace, sea, and land borders. The aim is to understand whether reported violations occur by accident and randomly or whether there are some systematic patterns regarding a geographical and temporal distribution. The focus will be on Russian violations since by far most events that are reported by the *Baltic News Service* (the main source of analysis) include Russia as the originator of violations. The second question of interest is whether Russian violations of the Baltic states' airspace can be explained as signals of resolve to deter the further expansion of NATO. The theoretical argument will draw on deterrence theory and signaling theory, followed by a description of violations of air and sea borders based on our own data collection of events that are reported in the *Baltic News Service*. After an assessment of the geographical distribution, the analysis will focus on variations in time and intends to assess whether Russia is more active militarily and violates borders more often in reaction to events surrounding NATO enlargement.

Minor military incidents as a signal of resolve

Russian foreign policy in the Baltic Sea region needs to be considered in the context of its relations with NATO and might best be explained by the concept of mutual deterrence. Militarized interstate disputes and other types of military action and border violations can serve as a credible threat and costly signal for Russia to demonstrate its resolve and to deter NATO activities and expansion. The deterrence literature peaked during the Cold War, focusing on relations between the United States and the Soviet Union involving nuclear as well as conventional deterrence. Deterrence theory, however, is not limited to this period and has continuously been developed in conceptual, theoretical, and empirical terms (Quackenbush 2011; Zagare 2006). In a survey article on conventional deterrence from a rationalist perspective, Paul Huth (1999) states that 'a policy of deterrence generally refers to threats of military retaliation directed by the leaders of one country to the leaders of another in an attempt to prevent the other country from resorting to the threat or use of military force in pursuit of its foreign policy goals.'[1] The threat to retaliate can be based on the threat to impose punishment or deny the opponent relevant gains or the attainment of its objectives, such as territorial expansion (Snyder 1960). Classical deterrence theory takes the form of a structural variant, emphasizing the importance of balance of power and second-strike capabilities as a stabilizing force, and a decision theory version allowing for cost–benefit calculations with different implications for accidental wars. In structural deterrence theory, accidental war is the only real threat, whereas in decision-making deterrence theory threats that leave something to chance as a way to raise the stakes can be part of the deterrence game (Quackenbush 2011; Zagare 2006).

Deterrence theory was swiftly extended to prevent attacks not only from the major powers directly but to include states from the respective blocs of the Warsaw Pact and NATO. Studies on extended deterrence more generally examined a defender's military, political, and economic ties with its protégé (Huth and Russett 1984; Huth 1988). Extended deterrence is still important and involves, for example, the US sending signals that it would punish attacks on NATO alliance partners by maintaining forces or sharing military equipment and training (Morgan 2018). This is of particular importance to the smaller Baltic countries and has even become more so in recent years, especially since the Ukraine crisis and deteriorating relations between Russia and NATO (Forsberg and Herd 2015). After a period of détente, nuclear disarmament, and close cooperation in the domain of security, weapon systems are once again being modernized and antagonism against the West is rising in Russia providing the concept of deterrence with renewed momentum, albeit in a somewhat different and more complex form in a global security environment (Morgan 2018).

A central challenge in deterrence theory, and international crisis behavior more generally, is to determine what makes a threat credible and how opponents can distinguish if a threat should be taken seriously or whether it is merely a bluff. Huth (1999) suggests various mechanisms that can help in identifying a credible threat, particularly the military balance in capabilities, reputation, and past behavior, interests that are at stake, and signals in a bargaining context. While information on military capabilities can help to identify whether states have the means for military confrontation, credible signals are necessary to learn if the opponent has sufficient resolve to actually use military means to the extent necessary. Threats must be taken seriously by the opponent, and, in the case of extended deterrence, there need to be credible

signals to allied and friendly states that the promised protection will be upheld if required (Morgan 2018). Past behavior can be informative in this context. A history of not honoring alliances and failure of intervention in the case that a protégé was attacked carries severe reputational costs for a state and weakens future alliance commitments (Morrow 2000; LeVeck and Narang 2017).

International military crises are situations with high levels of private information regarding military strength and uncertainty about the decision-makers' resolve to use military force. This, together with incentives to misrepresent relative power, makes it difficult to find a negotiated agreement. A leader's announcements about their foreign policy intentions can become more credible if rendered costly. Most of all, what is relevant is how the foreign policy statement is perceived by the opponent. Deterrence will only work if the potential attacker is convinced that the defender is determined to fulfil what they have threatened. Signals become costly if the leader faces significant *ex post* domestic audience costs if they backed down from the threat. Thus, an announced threat can tie a leader's hands if unsuccessfully managing a foreign policy crisis comes at the loss of domestic support. Alternatively, signals can carry *ex ante* costs through measures that significantly increase the risk of military conflict and carry financial costs, such as the mobilization of troops, building weapons, or conducting conflict on a limited scale. Such sunk-cost signals encompass costs regardless of the adversary's reaction (Fearon 1995, 1997).

Minor military incidents can be such a costly signal to demonstrate resolve and willingness to use military force and thus to deter the opponent from further steps in a crisis escalation. Portraying war as part and not as an alternative to a bargaining process, Wagner (2000) emphasizes that military operations do not necessarily aim to change the military capabilities of the opponent but can also be conducted to the effect of revealing information about relative capabilities and thus influence the bargaining behavior of the parties in conflict. Small-scale military incidents and minor border violations could be a way to send such costly signals to an opponent and deter them from further action.

Russia's increased level of external pressure has been linked to NATO's eastward enlargement by prominent scholars (for example, Mearsheimer 2014). Russian military activities in the Baltic Sea region have been frequently interpreted to follow deterrence logic, and the demonstration of military power and airspace violations could be expressions of such signals of resolve. In the case of Russia, the signals of resolve might not only be intended for NATO but also for neighboring countries that are still considered to be in Russia's sphere of influence and have expressed interest in moving under NATO's security umbrella. After the dissolution of the Soviet Union, Russia is struggling to maintain its power status globally as well as regionally. In an analysis of Russian–Ukrainian relations since 1991 Elias Götz (2016) argues that how major powers try to secure their position in the perceived region of influence and the instruments that they use for this aim depend on the level of external pressure. The level is high when smaller neighboring states attempt to cooperate with extra-regional great powers. In this case, the regional power will apply hard power tools including the threat or even use of military force. If smaller states are willing to cooperate with a regional great power, the latter will use soft power instruments to maintain and increase its influence (Götz 2016). Thus, Russia presumably demonstrates its military resolve more often if countries in its near abroad move closer to the Western security alliance.

The Baltic states and Poland's search for new partners in the West was, besides an interest in economic integration, to a considerable extent motivated by the aim to secure protection against Russia. The attempt to join NATO was a way to establish safeguards, especially for the small Baltic countries (Asmus 2002; Lane 1997; Jakniunaite 2016). Despite some reluctance and opposition, Russia accepted NATO's first wave of eastern extension, whereas the enlargement of NATO to the Baltic states in 2004 was more seriously protested and considered a threat by the Russian military and defense establishment. Relations became more strained when NATO considered the accession of Georgia and Ukraine in 2008 (Forsberg and Herd 2015).[2] Therefore, the variation in militarized incidents and border violations can be explained supposedly with developments that are related to the role of NATO and its enlargement. Russia is likely to increase its level of military activity where it has an interest in sending signals of strength and resolve to deter NATO from extending and intensifying its engagement in the region. The following assessment will examine whether more Russian airspace and border violations can be observed surrounding events related to NATO enlargement.

Geographic distribution of border and airspace violations in the Baltic Sea region

An inspection of widely used data collections reveals that few military incidents are reported for the Baltic Sea region. There are, for example, only six militarized interstate disputes (MID) in the Correlates of War data collection (Palmer et al. 2015), which took place in the Baltic Sea region between 1993 and 2010, and all include Russia as one conflict state in the dyad. While some incidents are related to the demarcation of borders, such as the incidents in the 1990s in Latvia, others, such as the airspace violations in Finland and Lithuania, involve elements that could be interpreted as Russian signals of resolve.[3] One incident involved 24 Russian helicopters armed with rockets that were conducting unauthorized flights over the Lithuanian capital Vilnius in March 1995. Generally, Lithuania allowed Russian military transit through its territory, including air force flights, but required special permission, for which Russia did not apply in this case. Parliamentary opposition groups in Lithuania viewed this air violation as 'an act of psychological pressure and intimidation of the Lithuanian public' (BNS 1995). Two additional MIDs are related to Russian planes that violated Finnish airspace in December 2007 and February 2008. Both incidents took place a few months after Finnish and Russian officials signed an agreement on limiting airspace violations (API 2007).

While the MID data coded events that involved a threat, display, or use of military force, including a wide spectrum of events, such as troop mobilizations or border violations (Jones, Bremer, and Singer 1996), this analysis will concentrate on a more specific type of event, namely violations of sovereign airspace and of territorial borders at land and sea. There is a relatively high number of close military encounters in the Baltic Sea region compared to other maritime regions (Åtland and Kabanenko 2020), and especially since 2014 overall military activity has increased; however, many of these almost 'routine' incidents go unreported in the media and are not considered to be dangerous. One way to establish whether an incident is serious and dangerous is to focus on 'military actions in breach of international law (for example, airspace and territorial waters' violations)' (Raynova and Kulesa 2018, 4). Thus, for the following

analysis only events with a specific violation of air, sea, or land borders of the Baltic Sea region littoral states were coded, excluding events that took place near a border or in international airspace.

The MID data's reliance on international news sources might be one reason for the small number of events that are reported for the Baltic Sea region. Additional incidents of a more minor nature might have occurred but are not reported in international news. Therefore, the following analysis draws on reports from the *Baltic News Wire Service*, which pays very close attention to any border violation involving Russia. The *Baltic News Service* is the leading news agency, which reports daily economic and political news from Estonia, Latvia, and Lithuania in five languages. Since the focus is on the Baltic states there might be a bias toward events that take place in Estonia, Latvia, and Lithuania and an underreporting of violations of the other littoral countries' airspace. Given the high level of threat perception in these countries, however, this news agency is strongly focused on border violations. The data collection on military incidents, which were reported in the *Baltic News Service* for the period between 1994 and 2018, was conducted in the English language version by using an automated search in *LexisNexis* based on a set of keywords. Two independent coders manually identified relevant events that contained specific violations of air, sea, or land borders and discrepancies were then verified in a group discussion.[4]

The data collection revealed a total of 50 violations of air, sea, or land borders and in 47 of these incidents Russia was the source of violation. Only three cases were identified when this was not the case. One incident occurred when a Latvian helicopter violated Lithuanian airspace in March 2001 while surveying an oil spill. In two other cases, Russia was the target of violation; however, both violations were of an accidental nature: In October 1996 a Swedish fighter jet crashed during a dummy attack on a Russian warship. Along similar lines, in January 1998, an Estonian aircraft flown by British pilots entered Russian airspace over the Kaliningrad region due to severe weather conditions and because they ran out of fuel as confirmed by Russian authorities after an investigation (BNS 1998).

The other 47 incidents in the database include Russia as the main border violator. The overwhelming majority of the incidents are airspace violations. The data collection revealed only one violation of a sea border by a Russian border patrol vessel that entered Lithuania's territorial waters during a turning maneuver, which took place in April 2018 (BNS 2018) and one violation of a land border in October 2003, involving Russian border guards in the pursuit of Lithuanian fishermen (BNS 2003). Therefore, the following analysis concentrates on airspace violations.

Table 1. Number of airspace violations by dyad, 1995–2018.

Violator	Target	Number of incidences
Russia	Estonia	35
Russia	Lithuania	9
Russia	Latvia	0
Russia	Finland	1
Latvia	Lithuania	1
Estonia	Russia	1
Sweden	Russia	1

As summarized in Table 1, the total of 45 reported Russian airspace violations are highly concentrated on two countries: Estonia and Lithuania. The *Baltic News Service* reports only one specific incident when Finnish airspace was violated in 2008. Instead, there are general references to violations of Finnish airspace, which are not specific enough to be included in the event data. The large bulk of events reported concerns 35 Estonian airspace violations from 1995 to 2018 as well as nine violations of Lithuanian airspace.

Strikingly, not a single report on a violation of the Latvian airspace has emerged. There was only one report of an event that happened between Russia and Latvia in October 1996. Three Russian navy ships entered Latvia's exclusive economic zone, which strictly speaking is not a violation of territorial waters (BNS 1996). Scrutinizing the Latvian language version of the *Baltic News Service* also did not reveal any additional violations for the Russian-Latvian dyad.[5] As Indulis Bērziņš, the Latvian ambassador to NATO, confirms in an interview in 2015, 'While identified Russian airplane maneuvers near our airspace are dangerous and unfriendly, Russia has never (not once) entered Latvian airspace' (BNS 2015b).

Some of the dyadic distribution and concentration on Estonia and Lithuania can be explained by geographic location. Russia's access to the Baltic Sea is limited to the region around St. Petersburg and the Kaliningrad region. As can be seen in Figure 1, Estonia and Finland have a long coastline along the Gulf of Finland that provides access to St. Petersburg, and Kaliningrad is located between Poland and Lithuania. There are frequent flights between St. Petersburg and Kaliningrad, the two main locations of the Russian Navy's Baltic Fleet (Nieto 2011).

Kaliningrad is of great geo-strategic, rather than economic, importance to Russia and is particularly perceived as threatening by the neighboring countries of Poland and Lithuania. This significance has been repeatedly highlighted with Russia using Kaliningrad as a site to store missiles and nuclear warheads and a base for repeated military exercises (Nieto 2011). A major incident, which triggered serious concern, occurred in September 2005 in Lithuanian airspace, when a Russian SU-27 fighter plane armed with four air-to-air missiles crashed in the Sakai district near the border with the Kaliningrad region. No one was killed and the pilot landed safely by parachute near the Jurbarkas district. During the flight, the pilot reported that he had lost his sense of direction, so the crash is believed to have been caused by a failure of the navigation system (BNS 2005c). A Lithuanian investigation committee later concluded that the violation happened because of a set of technical, organizational, and human factors (BNS 2008c). The incident, however, caused a large-scale debate about the security situation in Europe. The Estonian MEP Tunne Kelam made an official statement at the European Parliament that the Russian military flights over the Baltic Sea to Kaliningrad are a long-term challenge for European security and described Kaliningrad as 'the last vestige of the Cold War' (BNS 2005d).

By far most Russian violations are reported over Estonian airspace. In the middle of the narrow stretch of the Gulf of Finland, there is the Estonian island of Vaindloo, which has a thin tip that stretches out to the north and is close to international waters and the flight path of (Russian) airplanes. As a matter of fact, of the 35 Russian violations of Estonian airspace, 25 occurred near Vaindloo. With few exceptions, Russian aircraft had their transponders switched on and transmitted their flight plan ahead of time. Estonia repeatedly protested such violations and Russia claimed that the violations did not take place. Reactions from and within Estonia are mixed. For Estonian politician Marko

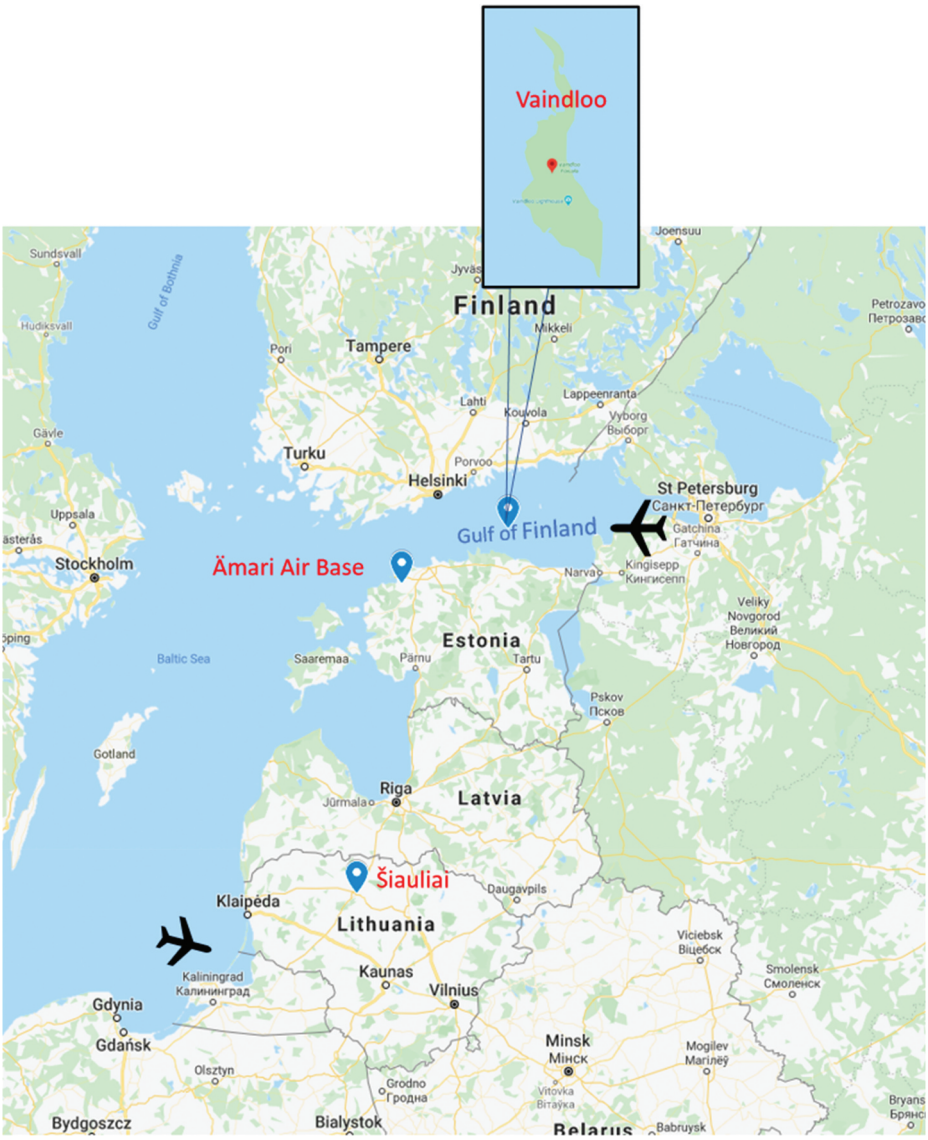

Figure 1. Map of the Gulf of Finland and the Eastern Baltic Sea. Source: Google, n.d.)

Mihkelson, 'with contemporary navigation equipment it is absolutely unthinkable that these entries are accidental acts of negligence' (BNS 2005a). Mihkelson declared in 2013 in light of another violation that 'The Vaindloo problem is not that it is a preferred destination for airspace violations, but the area is a relic of Soviet times, where Russian air traffic control overlaps with Estonian airspace' (ERR News).

The geographic distribution confirms that the violations are not purely random and only a few of the violations occurred by accident or because of severe weather conditions, such as the example of a pilot having lost orientation and crashed. Some of the geographic concentration of events can be explained by the narrow air corridor above

the Gulf of Finland and the difficulties of navigating it. The high concentration of events around Vaindloo is striking and suggests that Russian pilots take a deliberate shortcut rather than violating airspace as pure acts of provocation. Mixed motives, however, cannot be excluded. The following analysis of patterns over time allows us to gain further insight.

Temporal patterns: accidents or violations as signals of resolve?

Based on the theoretical arguments of the deterrence literature, we expect to see heightened military activity and airspace violations prior to NATO enlargements in 1999 in Poland, 2004 in the Baltic states, and possibly in 2008 when the NATO accession of Georgia and Ukraine was prominently discussed. Given the geographical patterns, the Russian violations of Estonian and Lithuanian airspace will be scrutinized more thoroughly. In Figure 2 the violations against Lithuania seem to be more randomly distributed, whereas the violations of Estonian airspace show that the distribution is not very even over time. Instead, there are heightened activities and clusters of violations around 2004 and 2014, the years surrounding NATO enlargement to the Baltic states and the annexation of Crimea.

In line with deterrence logic, Figure 2 provides some support for our expectation of more frequent demonstrations of Russian military force prior to the accession rounds of 1999 and especially of 2004. The first years after the disintegration of the Soviet Union, referred to as the 'honeymoon period' (Forsberg and Herd 2015, 43), involved only a few minor incidents in the Baltic Sea region, for example, the incident in 1995 that was

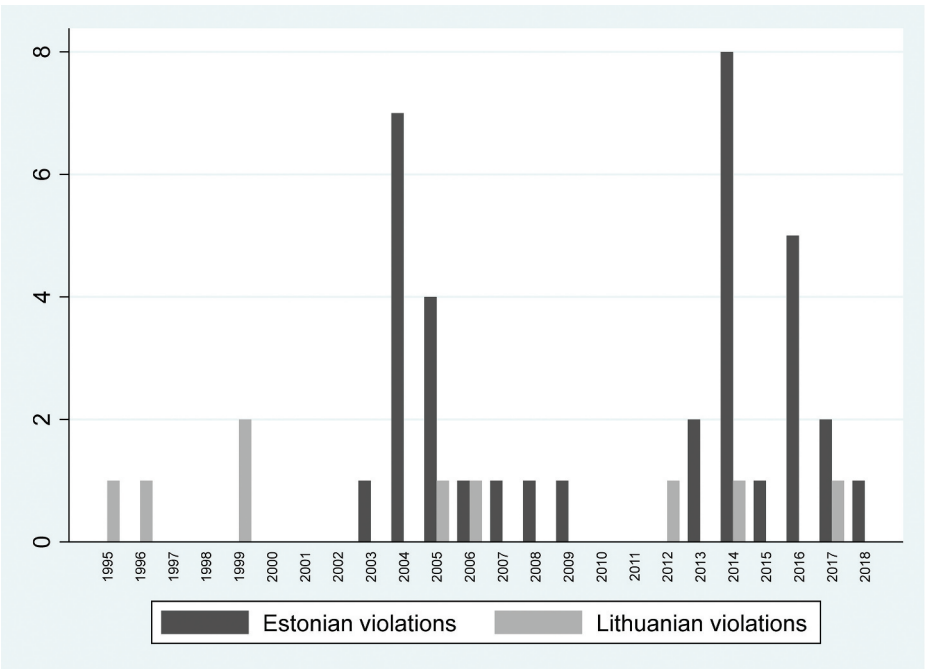

Figure 2. Russian airspace violations in Lithuania and Estonia over time.

criticized by the opposition, but presumably sanctioned by the Lithuanian government. Poland and the Baltic states developed an early interest in joining the Western military alliance. For the Baltic states, the request for NATO enlargement was motivated by the appeal to obtain protection from Russia, which was generally reluctant in withdrawing from, and demilitarizing, the Baltic states. Acknowledging some of the Russian concerns, the Baltic states were not part of the first wave of NATO accessions (Lane 1997). While Russia was concerned about the enlargement, its approach, although unsuccessful, was to seek legal guarantees (Forsberg and Herd 2015).

In March 1997, the NATO-Russia Founding Act was signed, which inter alia forbade the permanent deployment of foreign troops in former Warsaw Pact states. Russia did not engage in extensive deterrent behavior in the Baltic Sea region in the late 1990s and early 2000s. To show its discontent, also related to NATO's Kosovo campaign, if anything, Russia replied mostly by leaving cooperative forums, such as the Partnership for Peace (Forsberg and Herd 2015). The data on airspace violations does not reveal much heightened activity leading up to the 1999 enlargement round. One incident, however, a violation of Lithuanian airspace for 17 minutes in January 1999, can be explained in the context of the NATO eastward expansion (BNS 1999). Furthermore, in an interview in 2004, the Commander of the Polish Defense Forces compared the violations prior to the Baltic states' accession to NATO with several similar incidents preceding the accession of Poland (BNS 2004a). Overall, until the early 2000s the *Baltic News Service* reported only a few incidents of border violations. Instead, the early phase of Vladimir Putin's first term as Russian president was marked by renewed cooperation with the Western states, the creation of the NATO-Russia Council and cooperation in the fight against terrorism (Forsberg and Herd 2015).

NATO enlargement to officially include the Baltic states on 29 March 2004 was preceded by a degree of increased Russian military activity. Yet, these are individual events. According to Marko Michelson in an interview with the *Baltic News Service* in 2005, the most blatant incident happened in October 2003 in Estonia, when a Russian fighter aircraft entered Estonian airspace and remained inside it for approximately 20 minutes. According to Michelson this incident played a substantial role in the decision of several NATO members to contribute their fighter jets to the Baltic Air Policing deployment (BNS 2005a). The violation took place six months before Estonia officially joined NATO. A few days before the official accession a Russian cargo plane entered Estonian airspace close to the island of Vaindloo on 19 March 2004 for a few minutes. Presumably the violation was committed with the aim to test Estonia's ability to control its airspace and to see how far the country would take its protests. According to Vice Admiral Tarmo Kouts, it was 'an attempt to check how well Estonia is prepared for such incident and how we will react' (BNS 2004a). Moscow, on the other hand, rejected all accusations. While there are some, although not a substantially higher number of reports on airspace violations beforehand, an increasing number of violations were evident immediately after March 2004, when the three Baltic states officially became NATO members.

After NATO accession in 2004, there was an increase in reported airspace violations, committed by Russian military forces. The reasons for this increase, however, could be manifold. The Russian air force might have engaged more in testing Baltic reaction capabilities, but equally plausible is the view that detection might have improved with the presence of NATO and its military deployments in the Baltic states. After the three Baltic states joined NATO, the Alliance started air-policing missions

over the region (Marmei and White 2017). The Baltic Air Policing mission started at the end of March 2004 and was originally equipped with four fighter jets, which were stationed in Šiauliai, Lithuania. As none of the Baltic states originally possessed a fighter jet, 14 NATO members agreed to contribute their own jets to the Baltic Air Policing mission on a rotational basis. During each rotation, the use of four fighter jets was planned. After the Crimea crisis, an extra six US fighters were deployed to the mission, but this number was cut in 2016 (McNamara 2014). Today, the Mission is present in the northern Lithuanian city of Šiauliai, as well as in the Ämari Air Base in Harju County in Estonia.

Figure 2 reveals a clear increase in violations in this period. Some individual incidents in 2004 and 2005 can hardly be considered as acts of provocation, for example, when both sides agreed that a Russian fighter bomber violated the Lithuanian border, as in the case of the SU-27 because the pilot lost orientation and crashed (BNS 2005c). Some cases are related to faulty flight documentation. One incident refers to a Russian aerial survey plane that had previously informed Estonian authorities of its intentions and had international observers on board under the Treaty of Open Skies but did not wait for authorization (BNS 2004b). The large majority, namely 8 out of 12 cases in these 2 years, are related to violations near Vaindloo island. In most cases, the Russian aircraft had their transponders on and submitted prior flight plans and/or were in communication contact with the Estonian flight control center. The only exception is the incident on 30 June 2005, when the transponder was switched off. Reactions to these violations in Estonia were quite clear in that Estonia strongly protested. Finland also reported that in this period violations of its airspace increased (BNS 2005b).[6]

In 2007 and 2008, violations took place in Finland, which is not a NATO member country. Due to its postwar history and strong economic ties, Russia and Finland maintained a relatively close and friendly relationship. Finland has tried to avoid any confrontation over sensitive issues (Leonard and Popescu 2007). Despite the deteriorating relations between NATO and Russia, Finland and Sweden maintained a policy of military 'nonalignment' (Belkin, Mix, and Woehrel 2014). Nevertheless, the close economic ties between Finland and Russia and the strategy of political noninvolvement do not entirely protect Finland from the occurrence of militarized disputes with Russia.[7] In February 2007 Russia tested the world's largest nuclear icebreaker in the Gulf of Finland but, in violation of a treaty, did not inform Finland of the test (BNS 2007).[8] In January 2008, a Russian aircraft violated Finnish airspace near the southern city of Porvoo. Following the statement of then Russian Defense Minister Anatoli Serdyukov, the violations of Finnish airspace were attributed to technical problems in the navigation system, indicating the unintentional character of this event. As a result of trilateral talks between Russia, Finland, and Estonia, the parties agreed that military aircraft over the Gulf of Finland would be equipped with a Global Positioning System and the exchange of information on their flight plans (BNS 2008b).[9]

In the following years the number of reported violations declined again. There were no violations reported between July 2008 and February 2012. NATO's consideration of a possible extension to include Georgia and Ukraine and the Russian invasion in Georgia led to a suspension in military and political cooperation between Russia and NATO (Forsberg and Herd 2015, 49) but not to an increase in airspace violations in the Baltic Sea region. Nevertheless, for the three Baltic states, the war in Georgia was a reason to reconsider their defense policy. Since 2010 Estonia has spent the NATO target of 2% of its GDP annually on defense and Latvia and Lithuania managed to achieve this rate in

2018 (Foreign Policy 2019).[10] After the Georgian war, East European and Baltic countries tried to coordinate their foreign policy strategies and action plans. In October 2008, NATO began airspace surveillance exercises in the Baltic Sea region.[11]

The Georgian-Russian war has also revealed limitations and shortcomings of the Russian military in terms of outdated military equipment and slow reaction (Jakniunaite 2016; Renz and Thornton 2012). This revelation was an important catalyst for the start of military modernization in Russia. In the period between 2008 and 2012 the Russian armed forces underwent considerable structural change. The main aims of the reform were the transformation of the military from a conventional mobilization army to a permanently combat-ready force, as well as developing new weapons and mobile elements, which could be quickly delivered and deployed in conflict zones (Konyshev and Sergunin 2018).

A second cluster of events in Figure 2 coincides with the Ukrainian crisis and the following annexation of Crimea in 2014, which rapidly changed the security landscape in Europe. After a few failed attempts at integrating Ukraine into NATO, the new Ukrainian president, who was originally seen as a pro-Russian candidate, took steps to move closer to the West by starting negotiations with the EU over a free trade agreement, including anticipated closer collaboration in foreign and defense policy, which Russia found unacceptable. This was followed by protests and violent demonstrations against Russian interference in Ukraine. The rising instability in Kyiv led to the overthrow of President Viktor Yanukovych, and the subsequent escalation of conflict between Russia and Ukraine cumulated in the annexation of Crimea (Götz 2016).

The aggressive and unprecedented actions and its violation of international law became a turning point for most East European countries to revise their national security strategies. In the Baltic states, fear mounted that Russia might also seize territory in a land grab operation. While Russia was perceived as a constant threat to the Baltic states, after 2014, the security discourse became more intense, incorporating more prominently the components of territorial defense and the fear of a Russian invasion. Within this changed context, the threat perception increased because of higher Russian defense spending and military modernization, among others in Kaliningrad, as well as large military exercises in 2009 and 2013 close to the Baltic states that sent signals of readiness for action (Jakniunaite 2016; Rostoks 2018). In-depth interviews with Latvian foreign policy decision-makers revealed that they paid special attention to the development of military capabilities and military training exercises. The negative perception of Russia's intentions also increased for reasons not related to military factors, such as the decline of democracy in Russia and its willingness to break accepted international rules, with the annexation of Crimea considered as a prime example (Rostoks 2018). As a result of Russian aggression, NATO sent troops and tanks to the Baltic countries and considerably strengthened the Baltic Air Policing Mission (Blair 2015). After the annexation of Crimea, NATO first raised the number of fighter jets to 16 but cut this figure by half in 2016 because they were considered of little use (The Lithuania Tribune 2015).

In Figure 2 the number of Estonian airspace violations increased considerably up to eight violations in 2014, which presumably is related to the generally heightened activity of that year (BNS 2015a). In 2014–2015 NATO also observed the increased activity of Russian military aircraft in the airspace above the Baltic Sea. 'We've seen long-range strategic bombers flying, which are not usual visitors in our region. Those moves are unnecessarily provocative,' said Estonia's Defense Minister Sven Mikser at the time

(BNS 2014a). The official representative of NATO, Ben Hodges, interpreted the air violations in the Baltic region as a Russian aspiration to demonstrate its military capabilities (BNS 2014a). Russian military transport planes violated airspace around Vaindloo in half of the events in Figure 2 in the years 2014–2015 with their transponders on. There were, however, also reconnaissance planes that intruded into Baltic airspace, and these typically had their transponders switched off but reacted to radio contact. Notably, no other violation cases produced such unparalleled controversy and contrary interpretations between Russia and Estonia as the Vaindloo cases. The Russian side usually denied the violation and claimed to follow an approved route over neutral waters in strict conformity with international rules, whereas Estonia registered violations (BNS 2014b). Most violations, however, are not to be considered as dangerous actions.[12] In many cases, flight plans are filed ahead of time, and, for example, as acknowledged by the Lithuanian defense ministry, half of the registered violations between 2005 and 2008 were by civilian planes, and some of the military violations can be explained by errors in flight formalities. Often NATO planes in the context of the air-policing mission do not even take off if the violations are very short (less than a minute) (BNS 2009).

In 2015, the number of violations went down to two, whereas 2016 was another spike year with generally heightened military activity. Russia showed its military muscle in several ways. In October 2016, Russia started to deploy Iskander-M missile systems, which have a range of 500 km and can carry both conventional and nuclear warheads, in the Kaliningrad region, near the Lithuanian border. With this deployment, Russia claimed it was responding to the US military expansion in the Baltic Sea region (Reuters 2018). Prior to this deployment, there were reports of only two airspace violations, which took place near the Vaindloo island. In both cases, the transponders were switched on (BNS 2016a, 2016c). While not a violation of airspace, the *Baltic News Service* reported that at the beginning of October 2016, NATO jets scrambled 20 times to intercept Russian military planes and warships in airspace in international waters. There was a surge in NATO scrambling, but the scrambling in reaction to Russian military activity close to sovereign airspace did not increase (Raynova and Kulesa 2018).[13] The Lithuanian Defense Ministry attributed heightened Russian activity over the Baltic Sea to the military exercises conducted in the region (BNS 2016d; see also, Raynova and Kulesa 2018).

Conclusion

After a period of accommodation and cooperative relations between Russia and NATO, tensions in the Baltic Sea region have increased in recent years. The aim of this study was to investigate patterns in the Russian airspace violations and to determine whether Russia increasingly engaged in deterrent behavior, sending signals of its resolve to use military force and to deter further NATO expansion. To start with, a systematic and detailed assessment of violations of sovereign airspace and territorial land and sea borders underscores that the region is relatively peaceful, thus to some extent countering conventional media accounts and a widespread perception of Russian threats and violations. There are generally few incidents of serious and dangerous airspace or border violations that were reported in the *Baltic News Service*. Whether the Baltic Sea region is more peaceful compared to other maritime regions, such as the Black Sea, where Russia more forcefully pursues its interests, has not been investigated in this

article. There is an indication that the Baltic Sea region observes more military encounters than other regions, but these incidents have less potential to escalate and are rather related to the general heightening of military activity in recent years (Åtland and Kabanenko 2020; Raynova and Kulesa 2018).

This analysis of violations of air, sea, and land borders reveals a clustering of events surrounding the NATO enlargement round in 2004 and the Ukrainian and Crimean crises in 2014. This pattern can be taken as evidence that military incidents in the Baltic Sea region occur not by accident but in reaction to external developments. The expectations that Russia might have increased its military activity prior to the two NATO enlargement rounds to include Poland in 1999 and the Baltic states in 2004, however, have only been partially met. The data collection reveals that only individual border violations prior to each round of enlargement support the deterrent logic regarding NATO enlargement. Instead, an increase after the accession of the Baltic states could also be related to an increased activity of the Baltic Air Policing Mission and related action-reaction patterns between NATO and Russia.

The assessment of reports of airspace violations in the *Baltic News Service* also reveals many violations that occur around the island of Vaindloo, which is in the middle of the narrow stretch of the Gulf of Finland. These violations could be interpreted as a provocation but might rather be motivated by taking a shortcut and not flying around the long northern stretch of the island. Overall, heightened military activity after 2014 around the Baltic Sea as well as in other regions leads to more traffic generally and thus to more encounters. An avenue for further research could be the investigation of airspace violations in the context of military exercises. Border and airspace violations do not appear to be a clear-cut signaling device. Instead, military activities with higher sunk costs might be better suited to sending a costly signal of resolve. A systematic analysis of military exercises, together with a narrative of their scenarios, could provide more insights regarding the relevance of deterrence theory in the Baltic Sea region.

While the focus of the analysis was on Russian border violations against the Baltic states, the results revealed that the role of NATO needs more systematic evaluation in the region. NATO increased its presence considerably, partly in reaction to Russian advances and at the requests for protection from the Baltic states and Poland. Our analysis cannot yet determine whether Russia increased its activities in reaction to the greater NATO presence or whether the enhanced surveillance activities by NATO improved detection of violations. To determine this, more information is necessary on NATO's Baltic Air Policing Mission in terms of its size and the intensity of surveillance flights.

Despite the recent increase in military tensions, one should not conclude that the region is slipping toward a large-scale war. The end of the Cold War and the fall of the Iron Curtain opened avenues for cooperation in several ways. While the presence and commitment of NATO might have a deterrent effect, some of the relative peacefulness can also be explained by a mutually beneficial exchange. Various forms of cultural, economic, and political cooperation and exchange have security externalities and advance peaceful relations in the region and beyond, supporting Immanuel Kant's notion of a perpetual peace (Russett and Oneal 2001).

Notes

1. While general deterrence aims at preventing military crises and their escalation in the first place, successful immediate deterrence aims at keeping an already-stated threat or imminent start of military confrontation (and thus a failure of general deterrence) from escalating into a large-scale military confrontation (Huth 1999; Morgan 2018).
2. This decision was first announced during the Bucharest Summit in April 2008, where the US, with the agreement of the other alliance members, openly lobbied for Georgian and Ukrainian entry into NATO (Karagiannis 2013).
3. The MIDs in the early years related to tension between Russia and Latvia and an unsettled border dispute that was solved in 2007.
4. The first reports from the archive of the *Baltic News Service* were dated 1994–1995. A systematic search was conducted for reports that contained the following combination of key words: 'Russia AND (Estonia/Latvia/Lithuania/Finland/Germany/Poland/Sweden/Denmark) AND Baltic AND violation AND (air/sea/naval/border/land).' Manual coding of these reports helped to avoid the inclusion of multiple reports of the same event. Events that had a specific civilian character, such as violations by a sports aircraft or a sailing boat, were excluded, limiting the following analyses to violations by state institutions, such as military aircraft or border guards. Additionally, all combinations of dyads in the Baltic Sea region have been verified to see if there are any reports concerning violations within the region, which did not include Russia.
5. Moreover, in 2015 NATO officials reported a list of violations by Russian military aircraft trespassing in European airspace. According to those statistics for the period 2014–2015, there were also no violations of Latvian airspace (Shaikh and Rosenbaum 2015). Additionally, the data has been cross-checked against a list of close military encounters compiled by the European Leadership Network (Frear 2015), which confirmed the violations of Estonian and Lithuanian airspace and the lack of events for Latvian airspace. There was only one case of a violation of Swedish airspace in Frear's list that went undetected in our data collection.
6. Not many reports on specific events of violations in Finland in the *Baltic News Service*, however, were detected, which is likely the result of a reporting bias toward the Baltic states. In case of violations, Russia has typically apologized to Finland, but has not admitted violations or reacted to Estonian protests (BNS, 9 June 2008).
7. For the last few decades, Russia has been Finland's second largest trading partner after Germany. Moreover, Finland is very dependent on the Russian energy sector; in 2016, around 63% of primary energy in Finland was imported from Russia (Jääskeläinen et al. 2018).
8. According to a treaty signed in 1996, Russia must inform Finland of any movement of a nuclear-powered ship closer than 300 kilometers from the Finnish border.
9. In mid-November, General of the Army Vladimir Mikhailov, commander of the Russian Air Force, declared that the air force had been given instructions on how to fly along the Gulf of Finland to the Kaliningrad Region and back. The instructions were endorsed by the defense minister, and it is the main document for all Russian air force crews flying on that route (*Baltic News Service*, 7 December 2005).
10. Despite the increase in defense spending, the Baltic states will not be able to resist a Russian invasion without external protection. The 2016 annual study of the Rand Corporation concluded that without a deterrent the size of at least seven brigades and air support protecting eastern Europe, Russia could overrun the Baltic states in 60 hours (Shlapak and Johnson 2016).
11. Despite increasing tensions, cooperation between Russia and NATO continued. A few years after the Georgian-Russian war, NATO started 'Vigilant Skies 2011,' its first counterterrorism military airspace exercise with the Russian Federation.
12. A list of close military encounters from March 2014 to March 2015 considers most incidents as near routine (Frear 2015).
13. For NATO, the Baltic Air Policing deployment is a welcome occasion for training. Following the statement of the head of the outgoing German contingent, Lieutenant Colonel Matthias Fensterseifer, the response of real (or alpha) scrambles 'does not necessarily mean infringement, they may be carried out in connection to escorting of other aircrafts that come too close to the Baltic air space, according to the then head of the German contingent' (BNS 2012).

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